LAB 7: Fuzzing for Input Validation Bugs

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1. **python marathon_registration_parser.py run on the buggy module**

```
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7> python marathon_registration_parser.py
==== Running format_bib_name ====
Enter the name to print on the bib: Akanskha_PES1UG25CS801
Bib name: AKANSKHA_PES

==== Running parse_finish_time ====
Enter a finish time as H:MM:SS (e.g. 3:45:10): 3:45:10
Finish time in seconds: 13510

==== Running average_split ====
Enter your 5 km split times in minutes separated by spaces (e.g. 25 26 24): 25 26 24
Average split (minutes): 25.0
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7>
```

2. **pytest test_marathon_registration_parser.py with your completed tests failing on the buggy module**

```
C:\Users\AKANKSHA\AppData\Local\Programs\Python\Python310\lib\site-packages\hypothesis\internal\conjecture\datatree.py:700: AttributeError
---------- Hypothesis ----------
You can reproduce this failure by adding @seed(279279751624333004996951443524077604238) to this test, or by running pytest with --hypothesis-seed=279279751624333004996951443524077604
238.
=========================== short test summary info ============================
FAILED test_marathon_registration_parser.py::test_format_bib_name_no_crash - AttributeError: 'TreeNode' object has no attribute 'is_exhausted'
FAILED test_marathon_registration_parser.py::test_parse_finish_time_no_crash - AttributeError: 'TreeNode' object has no attribute 'is_exhausted'
FAILED test_marathon_registration_parser.py::test_average_split_no_crash - AttributeError: 'TreeNode' object has no attribute 'is_exhausted'
============================== 3 failed in 1.41s ===============================
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7>
```

3. **python marathon_registration_parser.py run on your fixed module**

```
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7> python marathon_registration_parser.py
==== Running format_bib_name ====
Enter the name to print on the bib: Akanksha_PES1UG25CS801
Bib name: AKANKSHA_PES

==== Running parse_finish_time ====
Enter a finish time as H:MM:SS (e.g. 3:45:10): 3:45:10
Finish time in seconds: 13510

==== Running average_split ====
Enter your 5 km split times in minutes separated by spaces (e.g. 25 26 24): 25 26 24 25 27
Average split (minutes): 25.4
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7>
```

4. **pytest test_marathon_registration_parser.py with all tests passing on your fixed module**

```
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7> python -m pytest test_marathon_registration
parser.py
==================== test session starts ====================
platform win32 -- Python 3.10.0, pytest-8.4.2, pluggy-1.6.0
rootdir: C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7
plugins: hypothesis-6.100.1
collected 3 items

test_marathon_registration_parser.py ...                [100%]

==================== 3 passed in 0.70s =====================
PS C:\Users\AKANKSHA\OneDrive\Documents\5th Sem\Software_Engineering\LAB-01\SE_LAB\SE_LAB\L6_L7_60_marathon_registration\LAB7>
```